

Nadim Mahmud

Applied Deep Learning | Robust and Explainable Machine Learning | AI for Environment and Agriculture
Dhaka, Bangladesh | mahmudnadim479@gmail.com | ORCID 0009-0002-9274-4850 | linkedin.com/in/nadim-mahmud-ml |
github.com/nadim7009

RESEARCH INTERESTS

I study how deep learning models behave when real data stops looking like the benchmark. My work so far covers robustness of image classifiers to blur, noise and lighting changes, segmentation of microplastic particles in electron microscopy images, and explainable models for environmental and agricultural data. I want to continue this in a master's or PhD programme, with a focus on reliable computer vision and explainable AI for environmental monitoring.

EDUCATION

Daffodil International University

B.Sc. in Software Engineering, major in Data Science

Dhaka, Bangladesh

Expected 2026

- CGPA: 3.73 / 4.00
- Thesis: Mitigating the correlation trap in email marketing through causal uplift modelling

PUBLICATIONS

All three papers were submitted in 2026 and are under peer review.

[1] Beyond clean accuracy: controlled robustness evaluation and corruption-aware training for plant disease classification

- **N. Mahmud**, M. S. Hossain, M. H. Riyad, J. J. Bushra, J. J. A. M. Sayor. Submitted to International Journal of Information Technology and Computer Science (IJITCS), MECS Press. Under review.

[2] Attention-enhanced deep learning for microplastic segmentation in scanning electron microscopy images

- **N. Mahmud** and co-authors. Submitted to the International Conference on Advances in Artificial Intelligence (AAI 2026), UniMAP, Malaysia. Under review.

[3] Explainable machine learning for individual carbon-footprint prediction from traditional, digital and AI behaviours

- **N. Mahmud** and co-authors. Submitted to an international conference. Under review.

RESEARCH EXPERIENCE

Robustness of plant disease classifiers under image corruption

2026

First author, paper [1]

- Designed a controlled benchmark: Gaussian noise, brightness reduction and Gaussian blur at three severity levels each, applied to the PlantVillage test set (54,305 images, 38 classes).
- Built a group-aware split that keeps all photos of one leaf on the same side, to stop near-duplicate images from inflating validation scores.
- Measured how much accuracy a strong EfficientNetV2-S baseline loses under each corruption and severity.
- Tested corruption-aware training of ResNet50 with each corruption alone and all three together, tracking clean, mean corrupted and worst-case accuracy.
- Compared four architectures and checked model reliability with Grad-CAM, reliability diagrams, expected calibration error and multiclass Brier score.

Attention-gated segmentation of microplastics in SEM images

2026

Paper [2]

- Trained a baseline U-Net and a four-gate Attention U-Net on 237 labelled SEM images with a fixed group-aware split (167 / 36 / 34).
- Compared the models on Dice, IoU, precision and recall to see where attention gates help and where they cost recall.
- Ran an ablation that removes each attention gate in turn, and robustness tests under noise and reduced brightness and contrast.

Explainable prediction of individual carbon footprints

2026

Paper [3]

- Built a leakage-safe scikit-learn pipeline for 18,000 synthetic respondents and 49 features covering housing, transport, diet, digital and AI use.
- Compared XGBoost, an MLP and Random Forest, and explained predictions with SHAP and permutation importance.
- Reported the limits of synthetic data openly and listed validation on real survey data as required future work.

- Undergraduate thesis: causal uplift modelling**
Daffodil International University

2026
- Comparing predictive response models with causal uplift models on the Hillstrom email marketing dataset under injected confounding, to measure how often correlation-based targeting picks the wrong customers.
- IrrigaSmart: soil-moisture risk prediction from field sensors**
Independent project, github.com/nadim7009/irrigasmart-soil-moisture-risk

2026
- Predicted 12-hour irrigation risk from 31,492 sensor observations on Belgian vegetable farms (KU Leuven open dataset), with a crop-cycle split, calibration analysis and sensor-fault stress tests. HistGradientBoosting reached F1 0.977 on unseen crop cycles against 0.972 for a threshold rule.

WORK EXPERIENCE

- Machine Learning Intern**
FlyRank AI

Jun 2026 – Present
Remote

TECHNICAL SKILLS

Languages and tools: Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter, Git, Google Colab, Kaggle
Deep learning: PyTorch, torchvision, CNNs, transfer learning, U-Net, Attention U-Net, ResNet, DenseNet, EfficientNet
Machine learning: scikit-learn, XGBoost, gradient boosting, Random Forest, cross-validation, uplift modelling
Evaluation and explainability: robustness benchmarking, calibration (ECE, Brier score), SHAP, Grad-CAM, permutation importance, ablation studies

CERTIFICATIONS

Data Science and Machine Learning with Python for Beginners, Ostad

LEADERSHIP AND SERVICE

- Media Executive, DIU Software Engineering Club**

Oct 2023 – Jan 2026
- Handled media and public relations for club events.
- Volunteer**

 - International Robotics Olympiad; Prothom Alo Math Olympiad; Prothom Alo Bondhushova

REFERENCES

Available on request.